SPHERE BRAWL: ROLLING RAMPAGE — ARENA TILE MATERIAL CORRECTION & FINALIZATION REPORT

Corrected Deliverable (62 Files + Root MANIFEST.json) | Visual Envelope: 11.80 x 2.00 x 11.80 Studs | Grid Pitch: 12.00 Studs (0.20-Stud Gap) | Top Surface: World Y = 51.0 | All 6 States < 250 Tris

Why Previous Imports Appeared Pale Grey in Roblox Studio (Source File vs. Studio Import Behavior): Inspection of the previous package confirmed that while the source `.mtl` (`Kd`/`Ke`) and `.glb` (`baseColorFactor`/`emissiveFactor`) contained the exact RGB palette, (1) Roblox Studio's `.obj` importer ignores `.mtl` `Kd`/`Ke` scalar colors when no texture map is bound, defaulting to *Medium stone grey (163, 162, 165)*; (2) the previous `.glb` stored all 4 material slots as primitives inside a single `mesh` node without `COLOR_0` vertex colors or `TEXCOORD_0` textures, which collapses into one grey `MeshPart` when merged; and (3) a single Roblox `MeshPart` supports only one `Color3` and one `Enum.Material` (`Metal` or `Neon`, never both). **How This Package Fixes It Without Altering Geometry:** Every `.glb` and `.obj` is now pre-split by material slot into 4 named child `MeshPart` nodes (*Part_MAT_DarkNeutral*, *Part_MAT_AltNeutral*, *Part_MAT_LightTrim*, *Part_EMI_**) AND includes baked `COLOR_0` vertex colors + an embedded 96×96 PNG palette atlas (`SphereBrawl_ArenaTile_PaletteAtlas.png`).

1. Exact Roblox Studio Material Setup Table (Child MeshPart -> Color3 & Enum.Material)

Material Slot	Imported Child MeshPart Name	Approved RGB (0-255) & Hex	Exact Roblox Color3 Value	Studio Material (Manual / Script)	Surface Role & What Must Be Set in Studio
MAT_DarkNeutral	**Part_MAT_DarkNeutral**	(32, 38, 54) #202636	Color3.fromRGB(32, 38, 54)	**Enum.Material.Metal**	Lower structural hull, corner bolsters, primary top armor pads. Set Material = Metal & Color3 in Studio (clear TextureID if using Part.Color).
MAT_AltNeutral	**Part_MAT_AltNeutral**	(42, 50, 70) #2A3246	Color3.fromRGB(42, 50, 70)	**Enum.Material.Metal**	Recessed interior deck floor (Y = +0.86 studs). Set Material = Metal & Color3 in Studio (clear TextureID if using Part.Color).
MAT_LightTrim	**Part_MAT_LightTrim**	(170, 190, 215) #AABED7	Color3.fromRGB(170, 190, 215)	**Enum.Material.Metal**	Chamfered upper perimeter frame (Y = +1.00 studs), inductor poles, and lock cleats. Set Material = Metal & Color3 in Studio (clear TextureID if using Part.Color).
EMI_Cyan	**Part_EMI_Cyan**	(0, 195, 255) #00C3FF	Color3.fromRGB(0, 195, 255)	**Enum.Material.Neon**	State 1 (Safe / Normal) restrained cyan seam inlays. Set Material = Neon & Color3 in Studio (emissive glow does not auto-convert).
EMI_Amber	**Part_EMI_Amber**	(255, 170, 0) #FFAA00	Color3.fromRGB(255, 170, 0)	**Enum.Material.Neon**	State 2 (Warning) amber caution perimeter and radial hazard stripes. Set Material = Neon & Color3 in Studio (emissive glow does not auto-convert).
EMI_Ice	**Part_EMI_Ice**	(0, 235, 255) #00EBFF	Color3.fromRGB(0, 235, 255)	**Enum.Material.Neon**	State 3 (Ice Corridor — active) longitudinal ice-blue cryo channels and frost bridges. Set Material = Neon & Color3 in Studio (emissive glow does not auto-convert).
EMI_Magnetic	**Part_EMI_Magnetic**	(165, 55, 255) #A537FF	Color3.fromRGB(165, 55, 255)	**Enum.Material.Neon**	State 4 (Magnetic Storm — active) concentric octagonal purple electromagnetic field rings. Set Material = Neon & Color3 in Studio (emissive glow does not auto-convert).
EMI_Kinetic	**Part_EMI_Kinetic**	(255, 125, 0) #FF7D00	Color3.fromRGB(255, 125, 0)	**Enum.Material.Neon**	State 5 (Boost Runway — active) forward (+Z) orange kinetic acceleration chevrons. Set Material = Neon & Color3 in Studio (emissive glow does not auto-convert).
EMI_DangerRed	**Part_EMI_DangerRed**	(255, 40, 40) #FF2828	Color3.fromRGB(255, 40, 40)	**Enum.Material.Neon**	State 6 (Ring Collapse — warning) danger-red perimeter bars and diagonal X-collapse motif. Set Material = Neon & Color3 in Studio (emissive glow does not auto-convert).

2. Preserved Geometry Envelope & 6-State Manifest Summary (< 250 Triangles Per Complete State)

State Name	Complete Mesh (.obj/.mtl/.glb)	Overlay Mesh (.obj/.mtl/.glb)	Tris (Base+Ovl)	Visual Bounds (Studs)	Child MeshPart Nodes in GLB / OBJ	Status
Shared Base	SM_ArenaTile_SharedBase	N/A (Reusable Base Chassis)	**84** (84+0)	11.80 × 2.00 × 11.80	Part_MAT_DarkNeutral, Part_MAT_AltNeutral, Part_MAT_LightTrim	**PASS**
1. Safe / Normal	SM_ArenaTile_01_SafeNormal	SM_ArenaTile_Overlay_01_SafeNormal	**214** (84+130)	11.8×2.0×11.8 (Ovl: 9.10×0.13×9.10)	Part_EMI_Cyan, Part_MAT_AltNeutral, Part_MAT_DarkNeutral, Part_MAT_LightTrim	**PASS**
2. Warning	SM_ArenaTile_02_Warning	SM_ArenaTile_Overlay_02_Warning	**224** (84+140)	11.8×2.0×11.8 (Ovl: 9.30×0.14×9.30)	Part_EMI_Amber, Part_MAT_AltNeutral, Part_MAT_DarkNeutral, Part_MAT_LightTrim	**PASS**
3. Ice Corridor — active	SM_ArenaTile_03_IceCorridor	SM_ArenaTile_Overlay_03_IceCorridor	**214** (84+130)	11.8×2.0×11.8 (Ovl: 7.24×0.13×9.60)	Part_EMI_Ice, Part_MAT_AltNeutral, Part_MAT_DarkNeutral, Part_MAT_LightTrim	**PASS**
4. Magnetic Storm — active	SM_ArenaTile_04_MagneticStorm	SM_ArenaTile_Overlay_04_MagneticStorm	**230** (84+146)	11.8×2.0×11.8 (Ovl: 9.10×0.13×9.10)	Part_EMI_Magnetic, Part_MAT_AltNeutral, Part_MAT_DarkNeutral, Part_MAT_LightTrim	**PASS**
5. Boost Runway — active	SM_ArenaTile_05_BoostRunway	SM_ArenaTile_Overlay_05_BoostRunway	**212** (84+128)	11.8×2.0×11.8 (Ovl: 8.90×0.14×8.90)	Part_EMI_Kinetic, Part_MAT_AltNeutral, Part_MAT_DarkNeutral, Part_MAT_LightTrim	**PASS**
6. Ring Collapse — warning	SM_ArenaTile_06_RingCollapse	SM_ArenaTile_Overlay_06_RingCollapse	**234** (84+150)	11.8×2.0×11.8 (Ovl: 11.20×0.14×11.20)	Part_EMI_DangerRed, Part_MAT_AltNeutral, Part_MAT_DarkNeutral, Part_MAT_LightTrim	**PASS**

3. Final File-Level Validation Report (Executed Sandbox Checks vs. External Application Checks)

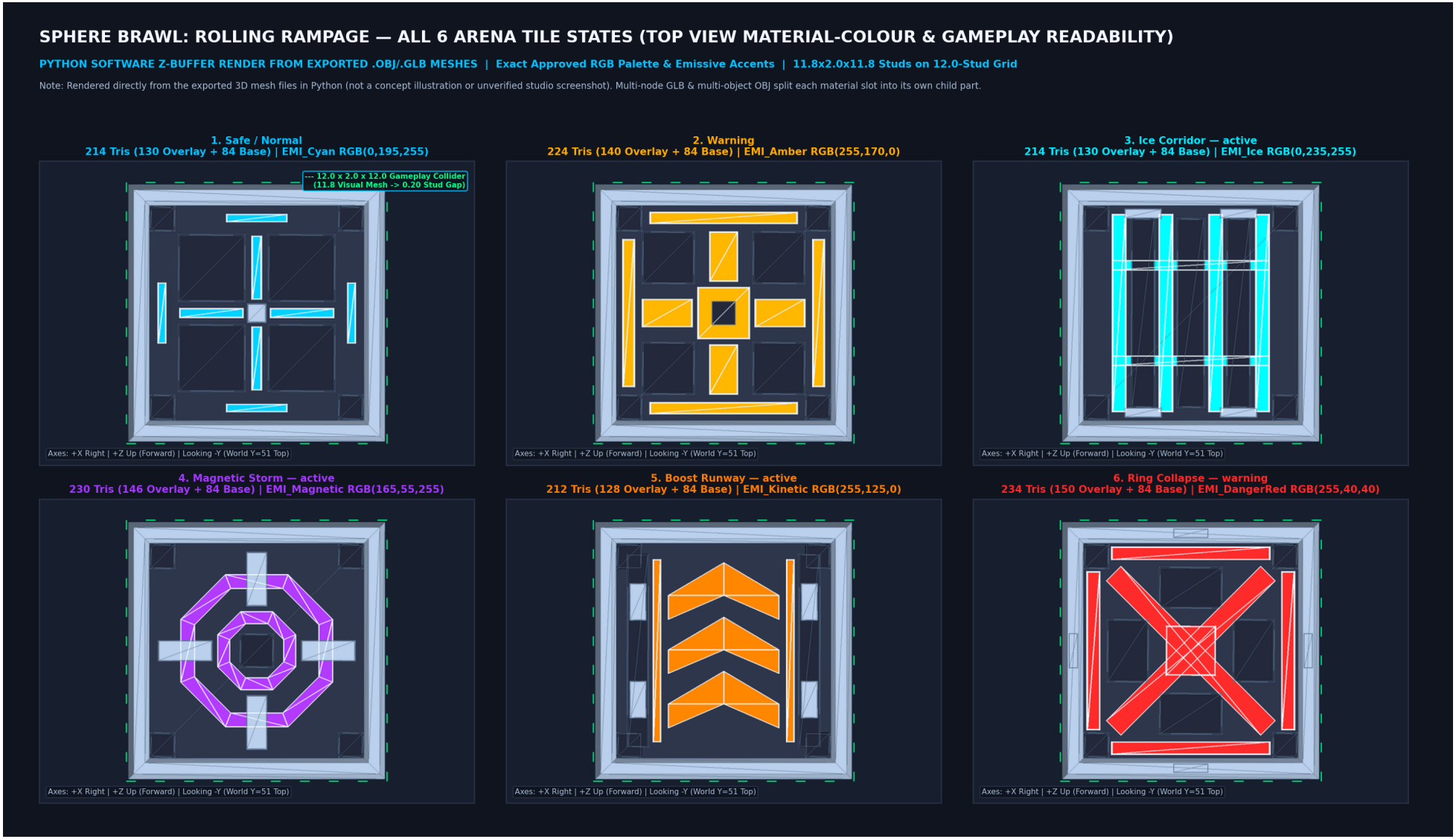
ID	Category	Validation Check	Validation Method	Measured Result	Status
EXEC-01	1. Executed in Sandbox	**OBJ Syntax, Face Indices, Object Subparts & Material References**	Independent Python line-by-line parser (`parse_and_validate_obj`) across all 13 `.obj` files (1 shared base + 6 overlays + 6 complete states)	0 syntax errors; 0 out-of-range `v`/`vt`/`vn` indices; 100% of faces have valid `v/vt/vn` triples, extended `v x y z r g b` vertex colors, and `o __` object grouping.	**PASS**
EXEC-02	1. Executed in Sandbox	**MTL References, RGB Definitions (`Kd`/`Ke`) & Atlas `map_Kd` Links**	Cross-checked every `.obj` `mtllib` and `usemtl` slot against its `.mtl` file and verified `Kd`/`Ke` RGB against the 9 approved palette entries (`parse_and_validate_mtl`)	13/13 `.obj` files resolve their local `.mtl`; max `Kd`/`Ke` deviation from approved RGB is < 0.0001; all 9 materials include `map_Kd SphereBrawl_ArenaTile_PaletteAtlas.png`.	**PASS**
EXEC-03	1. Executed in Sandbox	**GLB Structural Validity, Multi-Node Split, `COLOR_0` & Base-Colour Values**	Parsed 12-byte GLB header + JSON + BIN chunks across all 13 `.glb` files (`parse_and_validate_glb`), checking child `Part_*` nodes, `baseColorFactor`, `emissiveFactor`, `COLOR_0`, `TEXCOORD_0`, and embedded PNG	13/13 `.glb` files valid; each complete state exports 4 child nodes (`Part_MAT_DarkNeutral`, `Part_MAT_AltNeutral`, `Part_MAT_LightTrim`, `Part_EMI_*`), exact `baseColorFactor`/`emissiveFactor`, `COLOR_0` vertex colors, and embedded 96x96 PNG atlas.	**PASS**
EXEC-04	1. Executed in Sandbox	**Dimensions, Centered Origins, +Y Up Orientation & 0.2-Stud Grid Gap**	Measured bounding boxes, centers, top surface elevation (`Y = +1.00`), and 12.0-stud grid spacing across all 13 `.obj` and 13 `.glb` files	All 6 complete tiles & shared base measure exactly 11.8000 x 2.0000 x 11.8000 studs centered at `(0,0,0)` (`[-5.90,+5.90]` X/Z, `[-1.00,+1.00]` Y -> `World Y=51.0` when centered at `Y=50.0`), leaving an exact 0.2000-stud gap on a 12.0000-stud pitch.	**PASS**
EXEC-05	1. Executed in Sandbox	**Triangle Counts Below 250 Budget Across All 6 Complete States**	Counted triangles in both `.obj` and `.glb` for SharedBase (84 tris), 6 Overlays (128–150 tris), and 6 Complete States	State 1 = 214, State 2 = 224, State 3 = 214, State 4 = 230, State 5 = 212, State 6 = 234 tris. All 6 complete states are strictly < 250 triangles (100% match between `.obj` and `.glb`).	**PASS**
EXEC-06	1. Executed in Sandbox	**Degenerate Faces, Face Winding & Outward Normals**	Computed cross-product triangle area `0.5 * ||(p1-p0) x (p2-p0)||` and `dot(n_geom, vn)` across all 2,248 triangles in all 13 meshes	0 degenerate triangles (min triangle area = 0.006897 sq. studs > 1e-7); 0 face-winding / normal mismatches (`dot(n_geom, vn) == 1.0` on all faces).	**PASS**
EXEC-07	1. Executed in Sandbox	**Deterministic Generator Reproducibility (Python & Blender Collector Parity)**	Re-ran `generate_arena_tiles_pipeline.py` to `/tmp/det_test_dir` and compared SHA-256 hashes of `.obj`, `.mtl`, and `.glb`; also compared triangle & material groups against `generate_arena_tiles_blender.py`'s `DeterministicGeometryCollector`	100% identical SHA-256 hashes across repeated runs; `generate_arena_tiles_blender.py`'s standalone procedural collector produces the exact same 84 base tris and 128–150 overlay tris per state.	**PASS**
EXEC-08	1. Executed in Sandbox	**Preview Image Readability (Material-Colour, Neutral Clay, Side, Wireframe, Top)**	PIL `Image.verify()` + resolution/contrast audit (`validate_preview_png`) + visual inspection of all 12 PNG sheets (6 per-state 4-panel sheets + 6 master comparison sheets)	12/12 PNGs decoded cleanly (2400x1440 to 3000x1725 px, pixel std 28.4–47.2); neutral clay geometry sheet and material-colour sheets are crisp and distinguishable at small gameplay scale.	**PASS**
EXEC-09	1. Executed in Sandbox	**Manifest-to-File Consistency & ZIP Integrity**	Verified `zipfile.ZipFile.testzip()` CRC-32 check and bidirectional path, byte-size, and SHA-256 parity between `MANIFEST.json` and the final ZIP archive	0 corrupted ZIP entries; 100% match across all 62 files (23,149,688 bytes uncompressed, 20,162,724 bytes compressed).	**PASS**
EXT-01	2. Requires External App	**Blender (`bpy`) Execution of `generate_arena_tiles_blender.py`**	Checked Linux container for `blender` executable and `bpy` module; verified standalone `DeterministicGeometryCollector` in Python	Blender / `bpy` is not installed in this container. `generate_arena_tiles_blender.py` is self-contained and syntax/collector-verified in Python, but running its `bpy` FBX/GLB export requires a workstation with Blender 3.x/4.x.	**NOT RUN (Requires Blender)**
EXT-02	2. Requires External App	**Roblox Studio 3D Importer Live Verification of Multi-Node GLB/OBJ & Material Setup**	Requires interactive Roblox Studio 3D Importer (`Import As Model = true`, `Merge Meshes = false`) and viewport inspection	Not executed in this headless sandbox. After importing in Roblox Studio, verify that the 4 child `MeshPart`s (`Part_MAT_*`, `Part_EMI_*`) appear under the root `Model` and apply `Enum.Material.Metal` / `Enum.Material.Neon` + `Color3` per `docs/README_Arena_Tile_Collection.md`.	**NOT RUN (Requires Roblox Studio)**
EXT-03	2. Requires External App	**Optional Luau Helper (`Roblox_ArenaTile_Integration.lua`) Live Studio Execution**	Static audit of `ApplyApprovedMaterialsToImportedModel` and `MeasureVisualBounds` (no Roblox Studio runtime in sandbox)	Reviewed statically: assigns `Color3` and `Metal`/`Neon` to `Part_*` children without resizing meshes or mutating the authoritative `12x2x12` collider. Live execution requires Roblox Studio.	**NOT RUN (Requires Roblox Studio)**

4. Concise Change Log — What Changed & Why

Component	What Changed	Why It Was Needed
Multi-Node GLB Export	Exported 4 child nodes (`Part_MAT_DarkNeutral`, `Part_MAT_AltNeutral`, `Part_MAT_LightTrim`, `Part_EMI_*`) under root node + `COLOR_0` + `TEXCOORD_0` + embedded PNG atlas.	Prevents Roblox Studio from collapsing all 4 materials into a single pale-grey MeshPart; allows per-slot `Color3` and `Metal`/`Neon` assignment.
Multi-Object OBJ + MTL	Added `o __` object headers, `vt` UVs, `v x y z r g b` vertex colors, and `map_Kd SphereBrawl_ArenaTile_PaletteAtlas.png`.	Roblox Studio ignores `.mtl` `Kd`/`Ke` when no texture is bound; adding `o` sub-objects and `map_Kd` preserves both part splitting and swatch colors.
Standalone Blender Script	Rewrote `scripts/generate_arena_tiles_blender.py` with `DeterministicGeometryCollector` + Principled BSDF + `COLOR_0` vertex colors.	Ensures the Blender script deterministically recreates the exact geometry and materials from scratch without relying on pre-existing `.obj` files.
Dual-Mode Previews	Added `ArenaTiles_All6States_NeutralClay_GeometryComparison.png` and `ArenaTiles_All6States_SideComparison.png` alongside Material-Colour sheets.	Separates neutral-lighting geometry inspection from material-colour identity review and labels all renders accurately as Python software Z-buffer renders.

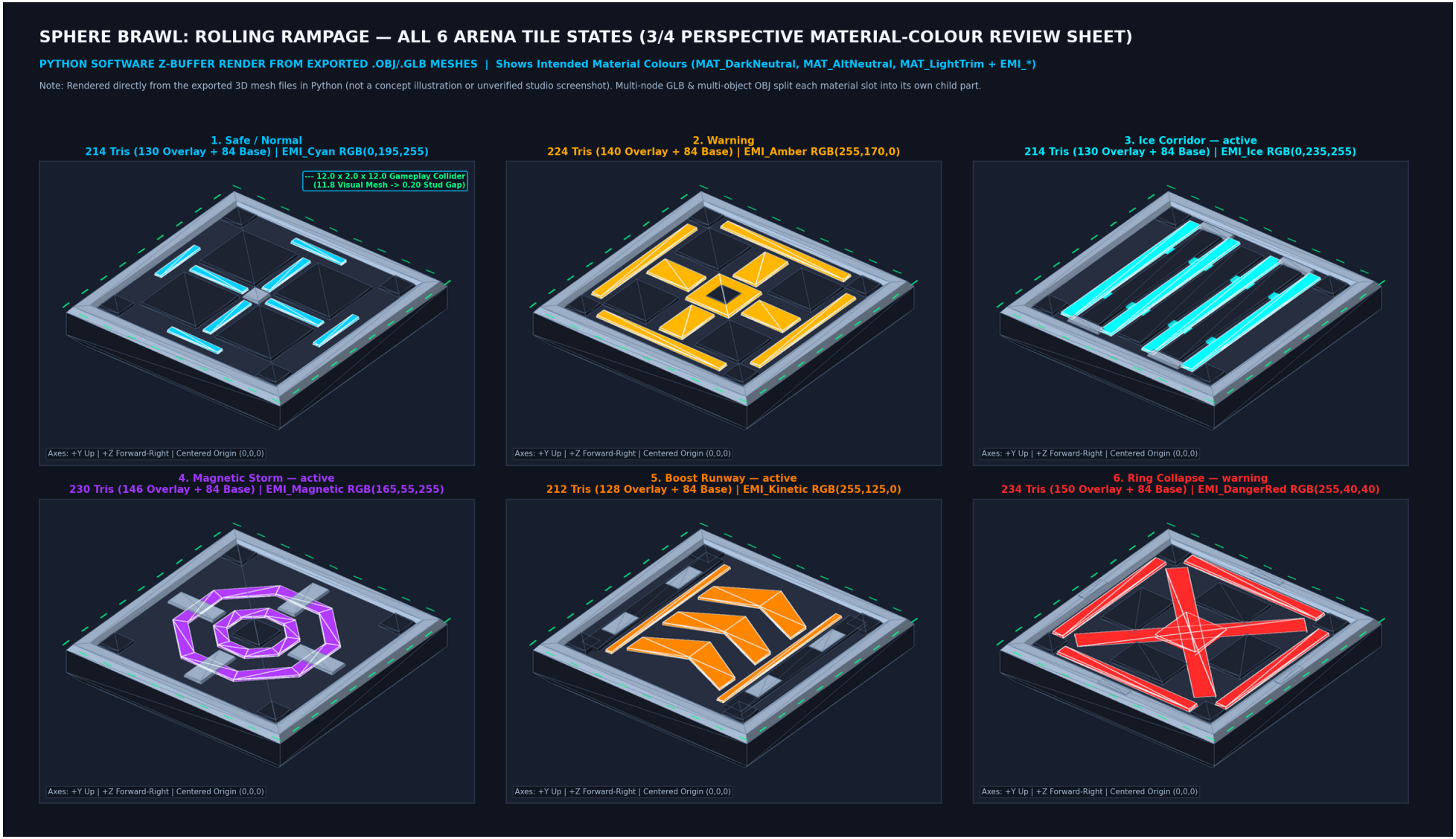
5. Material-Colour Review — All 6 States Top View (Gameplay Camera Readability)

Python software Z-buffer render directly from the exported `.obj`/`.glb` meshes showing the exact approved RGB palette (`MAT_DarkNeutral`, `MAT_AltNeutral`, `MAT_LightTrim`, and state `EMI_*` accents).



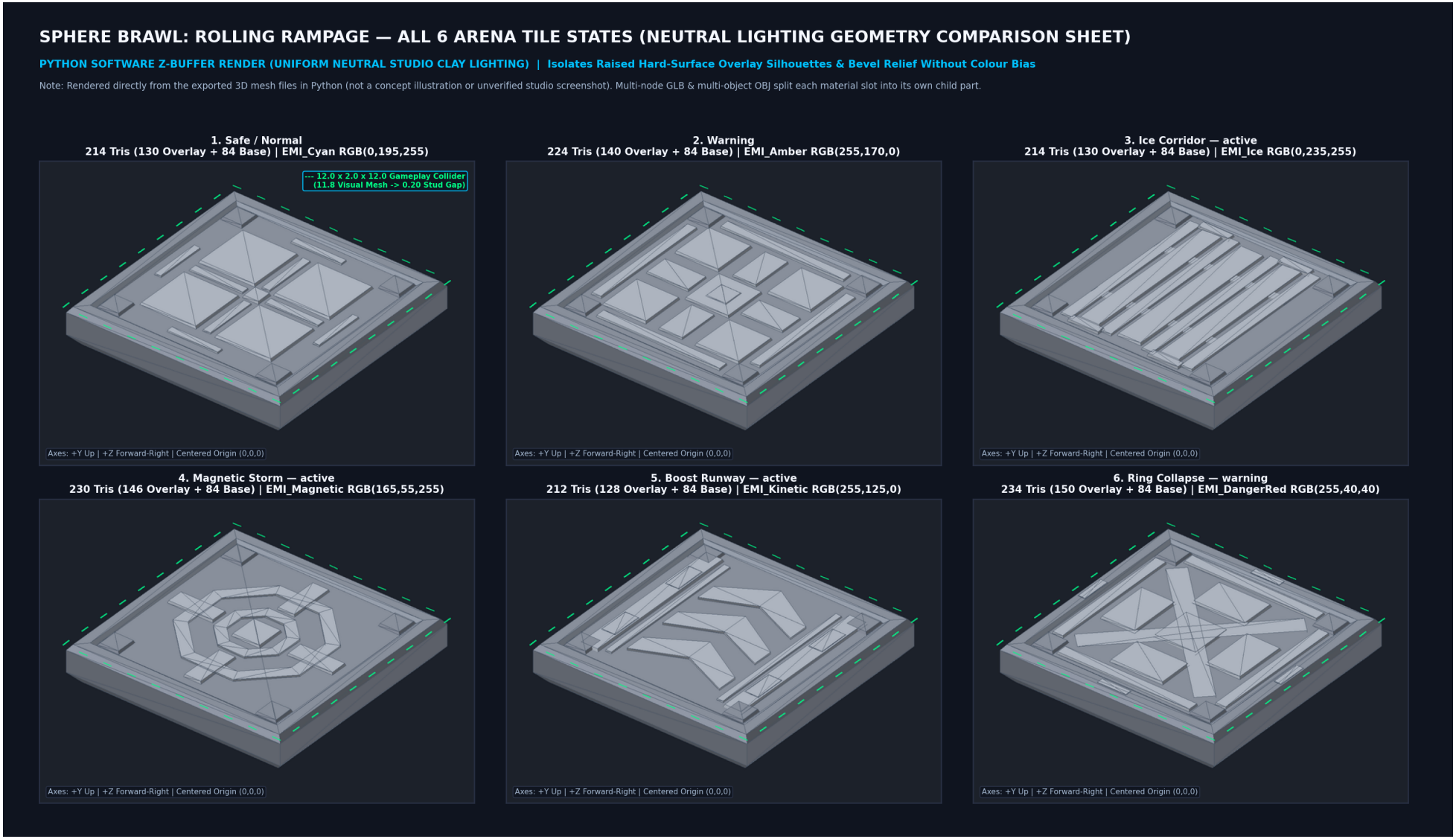
6. Material-Colour Review — All 6 States 3/4 Perspective View

Python software Z-buffer render showing 3/4 perspective bevels, dark metallic armor plates, light metallic trim, and restrained emissive hazard markings.

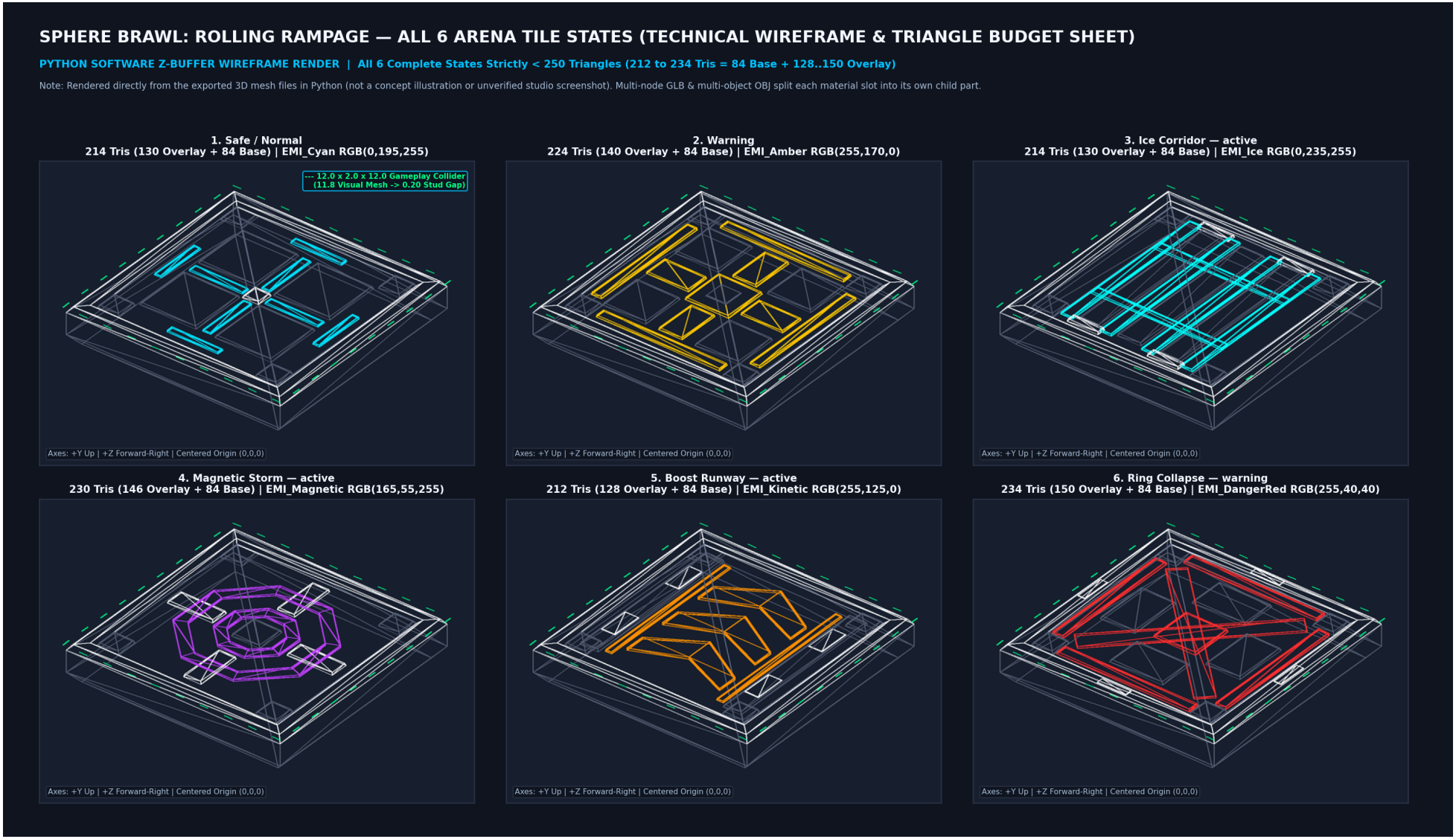


7. Neutral Lighting Geometry Comparison — All 6 States (Uniform Clay Shading)

Python software Z-buffer render under uniform neutral studio clay lighting to inspect raised hard-surface silhouettes, chamfers, and overlay relief without colour bias.



8. Side Elevation Profile & Technical Wireframe Comparison (< 250 Tris)



9. 2×3 Arena Grid Pitch & Small-Scale Gameplay Readability (12.0-Stud Pitch / 0.20-Stud Seam Gap)

